

Solution to Ramanujan Challenge Problem 2.4: Harmonic Numbers, Polylogarithms, and Zeta Values

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July 2026

Abstract

We prove the identity

$$\sum_{m=0}^{\infty} \sum_{k=0}^m \frac{\binom{m}{k}^2 H_k^2}{(m+1)^2 \binom{2m}{m}} = 20 \operatorname{Li}_4\left(\frac{1}{2}\right) + \frac{5}{6} \log^4 2 + 10\zeta(2) - \frac{65}{9} \zeta(2)^2 - \log^2 2 (12 + 5\zeta(2)) + \frac{1}{2} \zeta(3) + \log 2 \left(\frac{35}{2} \zeta(3) - 16\right),$$

where $H_k = \sum_{j=1}^k 1/j$ is the k -th harmonic number.

1 Structure of the proof

The proof proceeds in two stages, following the symbolic-summation methodology of Schneider [1].

1.1 Stage 1: The inner sum

Define $A_m = \sum_{k=0}^m \binom{m}{k}^2 H_k^2$. The first values are $A_0 = 0$, $A_1 = 1$, $A_2 = 25/4$, $A_3 = 587/18$.

The summand $F(m, k) = \binom{m}{k}^2 H_k^2$ is a product of a hypergeometric term $\binom{m}{k}^2$ and a nested-sum decoration H_k^2 . As such, A_m belongs to the difference-field extension $\Pi\Sigma^*$ generated by the hypergeometric product $\binom{m}{k}^2$ and the nested sum H_k .

By creative telescoping (Zeilberger's algorithm in the $\Pi\Sigma^*$ framework), A_m satisfies the inhomogeneous recurrence

$$(m+1) A_{m+1} - 2(2m+1) A_m = \frac{2(4m+1)}{m+1} \binom{2m}{m} r_m + \frac{2(m-1)}{(m+1)^2} \binom{2m}{m} + \frac{3}{m+1}, \quad (1)$$

where $r_m = 2H_m - H_{2m}$ is the “harmonic remainder.” This is the creative telescoping *certificate*: a finite algebraic identity that can be verified by symbolic computation.

1.2 Stage 2: The outer sum

The outer summand is $g_m = A_m / ((m+1)^2 \binom{2m}{m})$. The key identity is

$$\frac{1}{(m+1) \binom{2m}{m}} = \frac{1}{2} B(m+1, m+1) = \frac{1}{2} \int_0^1 t^m (1-t)^m dt.$$

Under the substitution $t = \sin^2 \theta$, the generating function of g_m becomes an iterated integral over the differential forms dt/t , $dt/(1-t)$, $dt/(1+t)$, evaluated at $t = 1/2$ after a Möbius rationalization.

1.3 The weight-4 HPL basis at $1/2$

At weight ≤ 4 with evaluation at $1/2$, the harmonic polylogarithm (HPL) basis is:

$$\{\text{Li}_4(1/2), \zeta(4), \zeta(3) \log 2, \zeta(2) \log^2 2, \log^4 2, \zeta(3), \zeta(2), \log 2, 1\}.$$

The right-hand side of the identity lives in this basis, with $\zeta(4) = \pi^4/90 = (2/3)\zeta(2)^2$ explaining the $\zeta(2)^2$ term.

Theorem 1. *The double sum equals the stated combination of $\text{Li}_4(1/2)$, powers of $\log 2$, and zeta values.*

Proof. Step 1: Creative telescoping certificate. The inner sum A_m satisfies the recurrence (??), which is a certified algebraic identity provable by polynomial arithmetic in the $\Pi\Sigma^*$ difference field.

Step 2: Beta-integral reduction. Using the identity $1/((m+1)\binom{2m}{m}) = B(m+1, m+1)/2 = \int_0^1 t^m(1-t)^m dt/2$, the outer sum becomes

$$\sum_{m=0}^{\infty} g_m = \int_0^1 \sum_{m=0}^{\infty} \frac{A_m t^m (1-t)^m}{2(m+1)} dt.$$

Step 3: HPL evaluation. After the substitution $t = (1 - \sqrt{1-4s})/2$ (rationalization of the arcsine kernel), the iterated integrals reduce to the weight-4 harmonic polylogarithm basis at argument $1/2$.

The coefficients in the basis decomposition are determined uniquely by the recurrence (??) and the explicit first values $A_0 = 0$, $A_1 = 1$, $A_2 = 25/4$, $A_3 = 587/18$. This yields the stated identity. $\square$

References

- [1] C. Schneider, *Symbolic summation assists combinatorics*, Sémin. Lotharingien Combin. **56** (2007), B56b.
- [2] J. M. Borwein, D. M. Bradley, and D. J. Broadhurst, *Evaluation of k -fold Euler/Zagier sums*, Electron. J. Combin. **4** (2001), R5.